The tools to prove consistency of Quine's NF

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Abstract

We give an account of the tools used in a recent proof of the consistency of TST+ (typed set theory with typical ambiguity). TST+ is equivalent to Quine's New Foundations; both are type theories in which the 'collection of all elements' is an element of itself, whose consistency has been an open problem. These tools include a new cumulative hierarchy model based on an innovative 'small support' condition, a new category generalising the category of nominal sets, some surprising counting lemmas, and interesting sets bijections. The application to the consistency proof is given in accessible detail, sufficient to reconstruct the full proof.

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1 Introduction

Consider the following *false* reasoning: define $x = \{a \mid a \notin a\}$. It is easy to check that $x \in x$ if and only if $x \notin x$. This is Russell's paradox and is one of the central paradoxes of naive set theory.

Zermelo-Fraenkel set theory (**ZF**) avoids paradox by insisting that a be *guarded*; we can only form $\{a{\in}y \mid a \notin a\}$ where y is already known to be a set. We pay the price that we cannot form 'reasonable' sets such as the **universal set** (the set of all sets) or the set of 'all sets with 2 elements'.[1]

$$\mathbf{univ} = \{a \mid \top\} \qquad \mathbf{2} = \{a \mid \exists b, c.(a = \{b, c\} \wedge b \neq c)\}$$

New Foundations (**NF**) avoids paradox by insisting on a *stratifiable* language [11]. In a stratifiable syntax, every variable and term can be assigned a *level* in $\mathbb{N}_{\geq 1}$ such that we only form $t{\in}s$ if $level(s) = level(t)$+1. So $a \in a$ and $a \notin a$ are outlawed because no matter how we assign levels, $level(a)$ can never be equal to $level(a)$+1.

The example expressions $\mathbf{univ}$ and $\mathbf{2}$ above are stratifiable, and are valid in NF. See the discussions in [1] and [8], and the clear summary with bibliography in [2]. One feature of NF is that it has a universal set (so the universe is an element, and is an element of itself), so NF does not need any hierarchy of universes as is often seen in foundations without a universal set.

If NF is obscure in computer science compared to HOL and ZF, this must be in part because it was not known consistent.[2] A proof of the consistency of Quine's NF (ConNF) using powerful generalisations of nominal techniques is presented in [4]. That paper, which is under review, is technical and aimed as it has to be at the reader expert in set theory and very interested in ConNF.

However, the proof contains several new ideas which I think may merit a broad audience. This paper provides an accessible introduction to these ideas, and highlights the tools more from a computer science (rather than a pure set theorist or NF-iste) point of view; see the list below. This paper started

[1] In ZF these are *proper classes*. A nice historical account of Russell's paradox is in [7]; for ZF see e.g. [9].

[2] Another criticism is that NF is inconsistent with the Axiom of Choice. But there is less to this than meets the eye: the NF elements that contradict Choice are ones that (like the universal set) the ZF universe is simply not concerned with, and an NF universe has a ZF-like initial segment that supports Choice. Thus, from an NF point of view it is not that ZF is consistent with Choice, rather that ZF is not even *not* consistent with it.

as a companion to the proof in [4] but has become an independent research contribution: most notably with the introduction of the category PART and the powerset object functor, and how this paper brings out the mathematical tools used in the proof in a new way. These tools include:

1. A generalisation of nominal 'finite support' to 'small partition support' (Definition 3.9).
2. A sets hierarchy with non-increasing cardinality (Definition 3.17; Theorem 3.18).
3. The category PART of uniformly partitioned sets (Definition 4.9).
4. The *power object* functor on PART (Definition 4.15), and the surprising fact that it preserves the first-order structure of sets (Theorem 4.23).
5. The technical but elegant 'squashing lemmas' of Subsection 4.5.

Our presentation is optimised for readability, while including enough information to reconstruct the technical details in [4]. The proof considered in this paper is of a theory TST+ (defined below), which is equiconsistent with NF by a result of Specker [13].

2 The language of stratified sets and TST+

2.1 Syntax and axioms

Definition 2.1. The language of TST+ is first-order classical logic with a binary predicate $\in$, and:

1. Each variable a has a level $level(a) \in \mathbb{N}_{\geq 1}$ and there are countably many variables of each level.
2. Predicates are subject to a **stratification** condition that if s and t are terms then $t \in s$ is only well-formed if $level(s) = level(t)$+1.

Example 2.2. Suppose $level(a) = 1$ and $level(b) = 2$ and $level(c) = 3$. Then $a \in a$ and $\forall a.\neg(a \in a)$ and $a \in c$ are not stratified, whereas $a \in b \wedge b \in c$ and $\forall a.\exists b.(a \in b)$ are stratified.

Definition 2.3. TST is a theory in the language of stratified sets with standard axioms for sets (see [4, Figure 2]), including extensionality, a **comprehension** axiom-scheme that for every (stratified) ϕ with just one free variable a, if b is not free in ϕ and $level(b) = level(a)$+1 then $\exists b.\forall a.(a \in b \Leftrightarrow \phi)$.

TST+ extends TST with **typical ambiguity** that for every closed ϕ (no free variables), we have

$$(\mathsf{TypicalAmbiguity}) \qquad \phi \Leftrightarrow \phi^+$$

where ϕ^+ is a copy of ϕ in which all variables are shifted up a level (so if $level(a) = 1$ and $level(b) = 2$ and $level(c) = 3$ then $(\forall a.\forall b.a \in b)^+ = \forall b.\forall c.b \in c$).

Remark 2.4. In condition 2 of Definition 2.1, terms s and t can only be variable symbols. While sufficient in principle, for usability we may prefer richer term languages. Two will interest us here:

- We may wish to permit constants from a model, once we have built it. See Definition 5.1.
- We may wish to permit explicit comprehension $\{a \mid \phi\}$. If so, we set $level(\{a \mid \phi\}) = level(a)$+1.

Explicit comprehension terms add no expressivity to TST+, because of the Comprehension axiom, but it is natural and convenient. I call this syntax the **language of stratified sets**: in [5] we study it from the point of view of rewriting theory and prove it confluent and strongly normalising — just like the simply typed λ-calculus, a syntax to which it is closely related, as we now discuss.

2.2 Connection to Higher-Order Logic

A connection between TST and Higher-Order Logic (**HOL**; simply-typed λ-calculus with base types *individuals* ι and *booleans* o) may be worth mentioning.

Consider TST with explicit comprehension terms and define a mapping from TST levels to HOL types, and TST terms and predicates to HOL terms, as follows:

- Define $hol(1) = \iota$ and $hol(i$+1$) = hol(i) \to o$ for $i \geq 1$.

- Biject level i variables of TST+ with type $hol(i)$ variables of HOL. For simplicity take this to be equality, so if $level(a) = i$ in TST+ then $a : hol(i)$ in HOL.
- Suppose $i \geq 1$ and $level(a) = i$ and ϕ is a TST+ predicate. Define $hol(\{a \mid \phi\}) = \lambda a.hol(\phi)$.
- Other connectives map unchanged, e.g. $hol(\forall a.\phi) = \forall a.hol(\phi)$ and $hol(\neg\phi) = \neg hol(\phi)$.

Under this translation the comprehension axiom $t \in \{a \mid \phi\} \Leftrightarrow \phi[a{:=}t]$ maps to β-equality $(\lambda a.s)t = s[a{:=}t]$, and the sets extensionality axiom maps to function extensionality.

Thus TST can be viewed naturally as a subsystem of HOL (this has not been noted in the NF literature, as far as I know). TST+ extends with *typical ambiguity*, which is not a HOL property. One way to model typical ambiguity would be to exhibit a bijection between the types $hol(i)$ and $hol(i{+}1) = hol(i) \to o$ for every $i \geq 1$; clearly this would suffice to prove $\phi \Leftrightarrow \phi^+$. Unfortunately, this is impossible by a diagonalisation argument, at least if we interpret $\to$ as 'all functions'.

So from the point of view of the translation above, the difficulty in proving the consistency of TST+/NF comes from finding a way to model typical ambiguity *without* running foul of diagonalisation. In a nutshell, we do this by restricting to small partition supported sets, then layering uniformly $\omega{+}1$-partitions on top of this restriction, and finally bijecting $hol(i{+}1)_\exists$ with $hol(i)_\exists$. Notation follows.

3 The hierarchy of sets with small partition support

We now build an extensional sets hierarchy model of TST+. We use 'small support' conditions to constrain the size of the hierarchy so that cardinality is the same at every stage ≥ 1 (Theorem 3.18). We need some machinery first; the impatient reader could jump to Definition 5.1 and work backwards.

3.1 Cardinalities

Definition 3.1. **1.** Write ON for the class of ordinals.
2. If X is a set write $\#X$ for the cardinality of X.
3. If $\lambda \in \mathsf{ON}$ is an ordinal then define a cardinality $\beth(\lambda)$ inductively by:
- $\beth(0) = \#\mathbb{N}$ where as usual $\mathbb{N} = \{0, 1, 2, \dots\}$ is the natural numbers.
- $\beth(\lambda{+}1) = 2^{\beth(\lambda)}$ (so if $\#\mathsf{X} = \beth(\lambda)$ then $(pow(\mathsf{X})) = \beth(\lambda{+}1)$).
- $\beth(\lambda) = \bigvee\{\beth(\lambda') \mid \lambda' < \lambda\}$ if λ is a limit ordinal.

Remark 3.2. $\beth(\omega)$ is the size of a model of Higher-Order Logic or Mac Lane set theory. In what follows, we will only need $\beth(0), \dots, \beth(\omega), \beth(\omega{+}1)$.

Notation 3.3. If $\alpha \in \mathsf{ON}$ is an ordinal then write $\aleph(\alpha)$ for the α-th cardinality above $\#\mathbb{N}$. So $\aleph(0) = \beth(0)$ and $\aleph(\alpha) \leq \beth(\alpha)$ for every $\alpha \in \mathsf{ON}$. The **generalised continuum hypothesis** (**GCH**) is the set-theoretic axiom that there are no 'extra' cardinalities:

$$(\mathsf{GCH}) \qquad \forall \alpha \in \mathsf{ON}.(\aleph(\alpha) = \beth(\alpha)).$$

Lemma 3.4. *Given GCH, there are* $\beth(0)$ *cardinalities below* $\beth(\omega{+}1)$*: namely* $\beth(0)$, $\beth(1)$, …, $\beth(\omega)$.

Definition 3.5. We assume GCH henceforth.

Remark 3.6. We can assume GCH because the assertion 'ConNF' is *arithmetical*, so by Schonfield's absoluteness theorem [12] 'GCH⇒ConNF' implies 'ConNF'.

Notation 3.7. Suppose X is a set.
1. If $\#X < \beth(\omega)$ then call X **small**.
2. If $\#X = \beth(\omega{+}1)$ then call X **large**.
We do not have a special name for a set with cardinality equal to $\beth(\omega)$, though one such will be important: the set of atoms $\mathbb{A}$ fixed in Definition 3.17.

3.2 Partitions and support

DEFINITION 3.8. Suppose A is a set.

1. Write $part(\mathsf{A})$ for the **partitions** of A. So $E \in part(\mathsf{A})$ is a subset $E \subseteq pow(\mathsf{A})$ that partitions A.
2. Call $P \in part(\mathsf{A})$ **small** when $\#P < \beth(\omega)$, and define $part_s(\mathsf{A})$ the **set of small partitions** by

$$part_s(\mathsf{A}) = \{P \in part(\mathsf{A}) \mid \#P < \beth(\omega)\}.$$

DEFINITION 3.9. Suppose X is a set with a permutation action over A.

1. Suppose $x \in \mathsf{X}$ and $E \in part(\mathsf{A})$. Write $E\$x$ and say that E **supports** x when

$$\forall \pi \in Perm(\mathsf{A}).\pi \in stab(E) \Rightarrow \pi{\cdot}x = x.$$

2. If every $x \in \mathsf{X}$ is supported by some *small* partition $E \in part_s(\mathsf{A})$ then say X has **small partition support** (over A). In symbols:

$$\forall x \in \mathsf{X}.\exists E \in part_s(\mathsf{A}).E\$x.$$

REMARK 3.10. Small partition support generalises a *small support* property that x be supported by a small $A \subseteq \mathsf{A}$ (itself a generalisation of the nominal 'finite support' condition from [6]):

- A small set $A \subseteq \mathsf{A}$ determines a small partition $\{\{a\} \mid a \in A\} \cup \{\mathsf{A} \setminus A\}$.
- Conversely if we start from small partition support and insist that all but one of the equivalence classes of E be small, then we recover a small support condition.

3.2.1 Representing powersets

DEFINITION 3.11. Suppose X is a set with a permutation action over a set A (it may be that $\mathsf{A} = \mathsf{X}$). Given $X \subseteq \mathsf{X}$ and a permutation $\pi \in Perm(\mathsf{X})$, define the **pointwise** permutation action $\pi{\cdot}X$ by

$$\pi{\cdot}X = \{\pi{\cdot}x \mid x \in X\}.$$

DEFINITION 3.12. Suppose X is a set and $\#\mathsf{X} \leq \beth(\omega{+}1)$, and $P \in part(\mathsf{X})$ is a small partition of X, and $X \subseteq \mathsf{X}$ (so X is a subset of X). Then:

1. Define a **representation map** $r_P(X)$ taking each $p \in P$ to a pair of cardinalities as follows

$$r_P(X)(p) = (\#(p \cap X), \#(p \setminus X)).$$

2. Given $X, X' \subseteq \mathsf{X}$, write $X \sim_P X'$ when there exists $\pi \in stab(P)$ such that $X' = \pi{\cdot}X$.

REMARK 3.13. So r_P maps $pow(\mathsf{X})$ to maps from equivalence classes in P to pairs of cardinalities no greater than $\beth(\omega{+}1)$. Slightly abusing notation we can write:

$$r_P \in pow(\mathsf{X}) \to P \to (\omega{+}1)^2$$

PROPOSITION 3.14. *Suppose X is a set, and fix some small partition $P \in part_s(\mathsf{X})$. Then:*

1. *If $X, X' \subseteq \mathsf{X}$ and $P\$X, X'$ then*

$$X \sim_P X' \quad \textit{if and only if} \quad r_P(X) = r_P(X').$$

2. *The set of possible values for $r_P(X)$ is small (has cardinality $< \beth(\omega)$).*
3. *Suppose $\mathcal{X} \in pow(pow(\mathsf{X}))$ and $P\$\mathcal{X}$. Then $X \in \mathcal{X}$ if and only if $r_P(X) \in \{r_P(X) \mid X \in \mathcal{X}\}$.*
4. *$\{X \subseteq pow(\mathsf{X}) \mid P\$X\}$ is small.*

Proof. 1. ▪ Suppose $X \sim_P X'$, so there exists $\pi \in stab(P)$ such that $X' = \pi{\cdot}X$. Because $\pi \in stab(P)$, for each equivalence class $p \in P$, π restricts to a bijection π_p on p. It is then a fact of Definition 3.12(1) (since bijections preserve cardinality) that $r_P(X) = r_P(X')$.

▪ Suppose $r_P(X) = r_P(X')$, so for every $p \in P$ there is a bijection π_p of $p \subseteq \mathsf{X}$ such that

$$\pi_p{\cdot}(p \cap X) = p \cap X' \quad \text{and} \quad \pi_p{\cdot}(p \setminus X) = p \setminus X'.$$

Combining these bijections, we obtain a bijection π of X such that $\pi{\cdot}X = X'$. By construction $\pi \in stab(P)$ and it follows that $X \sim_P X'$ as required.

2. The domain of $r_P(X)$ is the set of equivalence classes in P, which is small by assumption. The range of $r_P(X)$ is the set of (pairs of) cardinalities no greater than $\beth(\omega{+}1)$, which by our assumption of GCH Lemma 3.4 has cardinality $\beth(0)$ and so is also small.[3] The result follows.
3. Unpacking definitions, $E\$\mathcal{X}$ means precisely that $X \in \mathcal{X}$ precisely when $[X]_E = \{\pi{\cdot}X \mid \pi \in stab(E)\} \subseteq \mathcal{X}$. The result follows.
4. From the above, noting that the powerset of a small set is itself small. ◀

3.2.2 Representing equivalence classes

Definition 3.15 and Proposition 3.16 are exact replays of Definition 3.12 and Proposition 3.14, only acting on $pow(\mathsf{X})^2 = pow(\mathsf{X}) \times pow(\mathsf{X})$ instead of on $pow(\mathsf{X})$. We outline the argument:

Definition 3.15. Suppose X is a set and $\#\mathsf{X} \leq \beth(\omega{+}1)$, and $P \in part_s(\mathsf{X})$ is a small partition of X, and $X_1, X_2 \in pow(\mathsf{X})$ (so X and X' are subsets of X). Then:

1. Define a **representation map** $r_P(X_1, X_2)$ taking each $p \in P$ to a 4-tuple of cardinalities by

$$r_P(X)(p) = (\#(p \cap X_1 \cap X_2), \#((p \cap X_1) \setminus X_2), \#((p \cap X_2) \setminus X_1), \#(p \setminus (X_1 \cup X_1))).$$

2. Given $(X_1, X_2), (X_1', X_2') \in pow(\mathsf{X})^2$, write $(X_1, X_2) \sim_P (X_1', X_2)$ when there exists $\pi \in stab(P)$ such that $X_1' = \pi{\cdot}X_1$ and $X_2' = \pi{\cdot}X_2$.

Proposition 3.16. *Suppose X is a set, and fix some small partition $P \in part_s(\mathsf{X})$. Then:*

1. *If $(X_1, X_2), (X_1', X_2') \in pow(\mathsf{X})^2$ and $P\$X_1, X_2, X_1', X_2'$ then*

$$(X_1, X_2) \sim_P (X_1', X_2') \quad \textit{if and only if} \quad r_P(X_1, X_2) = r_P(X_1', X_2').$$

2. *The number of possible values for $r_P(X_1, X_2)$ is small (has cardinality $< \beth(\omega)$).*
3. *Suppose $\mathcal{X} \in pow(pow(\mathsf{X})^2)$ and $P\$\mathcal{X}$. Then $(X_1, X_2) \in \mathcal{X}$ if and only if $r_P(X_1, X_2) \in \{r_P(X_1, X_2) \mid X \in \mathcal{X}\}$.*
4. *$pow_{\$P}(pow(\mathsf{X})^2)$ is small.*

Proof. Exactly as the proof of Proposition 3.14, but using the representation from Definition 3.15. ◀

3.3 The hierarchy of universes

We will see in Theorem 5.7 that Definition 3.17 ([4, Definition 4.3]) specifies a model of TST+.

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[3] Without GCH Lemma 3.4 might fail, because there might be $\beth(\omega{+}1)$ (instead of $\beth(0)$) many cardinalities less than or equal to $\beth(\omega{+}1)$, which would increase the cardinality of the image of r_P. We really only need the set of cardinalities less than or equal to $\beth(\omega{+}1)$ to have cardinality strictly less than $\beth(\omega)$. In a sense given in Remark 3.6, this does not matter very much.

DEFINITION 3.17. Fix a set of **atoms** $\mathbb{A}$ such that $\#\mathbb{A} = \beth(\omega)$. Then define

$$\begin{array}{ll} \mathsf{V}_0 = \mathbb{A} & \mathsf{V}_{i+1} = \{X \subseteq \mathsf{V}_i \mid \forall 0 \leq j \leq i. \exists P_j \in \mathsf{EQ}_j. P_j \$ X\} \\ \mathsf{EQ}_0 = part_s(\mathsf{V}_0) & \mathsf{EQ}_{i+1} = \{P \in part_s(\mathsf{V}_{i+1}) \mid \forall 0 \leq j \leq i. \exists P_j \in \mathsf{EQ}_j. P_j \$ P\} \end{array}$$

As promised at the start of this Section, the hierarchy does not increase in size beyond level 1:

THEOREM 3.18. **1.** $\#\mathsf{V}_0 = \beth(\omega)$.
2. $\#\mathsf{V}_{i+1} = \beth(\omega{+}1)$ *for* $i \geq 0$.
3. $\#\mathsf{EQ}_i = \beth(\omega{+}1)$ *for* $i \geq 0$.

Proof. We consider the sets involved:

- By Definition 3.17 $\#\mathsf{V}_0 = \beth(\omega)$ and $\#\mathsf{EQ}_0 = \beth(\omega{+}1)$.
- The support condition in Definition 3.17 is trivial for the case of V_1 because $X \subseteq \mathbb{A}$ is supported by the small partition $P_0 = \{X, \mathbb{A} \setminus X\}$. So $\mathsf{V}_1 = pow(\mathsf{V}_0)$ and $\#\mathsf{V}_1 = \beth(\omega{+}1)$.
- It follows from Proposition 3.16 that $\#\mathsf{EQ}_1 = \beth(\omega{+}1)$, because there are $\beth(\omega{+}1)$ many $E_0 \in \mathsf{EQ}_0$ and a $\beth(\omega{+}1)$-sized union of small sets has cardinality $\beth(\omega{+}1)$.
- It follows from Proposition 3.14 that $\#\mathsf{V}_2 = \beth(\omega{+}1)$, because there are $\beth(\omega{+}1)$ many $E_0 \in \mathsf{EQ}_0$ and a $\beth(\omega{+}1)$-sized union of small sets has cardinality $\beth(\omega{+}1)$.
- The cases of EQ_i and V_{i+2} for $i \geq 2$ are just as the previous two cases. For instance, $\#\mathsf{V}_3 = \beth(\omega{+}1)$ because there are $\beth(\omega{+}1)$ many possible $E_1 \in \mathsf{EQ}_1$, and so on. ◀

REMARK 3.19. **1.** Amazingly, the hierarchy above satisfies stratified comprehension: though cardinality does not increase we can interpret comprehension over stratified predicates. The proof is not even very hard: see Proposition 5.5. Also amazingly, we have Typical Ambiguity, so the hierarchy above models TST+. That proof is much harder and requires the machinery of the next Sections.
2. Theorem 3.18 just needs every $X \in \mathsf{V}_{i+2}$ and $P \in \mathsf{EQ}_{i+1}$ be supported by some small partition $P_i \in \mathsf{EQ}_i$. The support condition in Definition 3.17 is stronger, that $X \in \mathsf{V}_{i+2}$ and $P \in \mathsf{EQ}_{i+1}$ be supported by small partitions for *every* lower level. The stronger condition is required for sets comprehension (see Remark 5.6).
3. Support conditions exist at every level and not just at the level of $\mathbb{A}$. $\mathbb{A}$ is a base case, but it is not the *only* set whose permutations and support conditions interest us. In particular, when we consider $\pi \in Perm(\mathsf{V}_i)$ there is no assumption in general that π be generated by some $\pi' \in Perm(\mathbb{A})$.

4 Support and uniformly partitioned sets

We now need to prove that Definition 3.17 is a model of TST+. Theorem 3.18 was a start, but we cannot just extend the bijections pointwise up the hierarchy; we would lose small partition support. We will develop something more sophisticated, culminating with the power object functor (Definition 4.15).

4.1 Being supported by a set

DEFINITION 4.1. **1.** Suppose A is a set and $A \subseteq \mathsf{A}$. Define $fix(A)$ and $moves(A)$ by

$$\begin{array}{r} fix(A) = \{\pi \in Perm(\mathsf{A}) \mid \forall a \in A. \pi(a) = a\} \\ moves(A) = fix(\mathsf{A} \setminus A) = \{\pi \in Perm(\mathsf{A}) \mid \forall a \in \mathsf{A}. \pi(a) \neq a \Rightarrow a \in A\}. \end{array}$$

2. Suppose X is a set with a permutation action over A and $A \subseteq \mathsf{A}$ and $x \in \mathsf{X}$. Define $A\$x$ and say that the set A **supports** x by

$$A\$x \quad \text{when} \quad \forall \pi \in Perm(\mathsf{A}). \big(\pi \in fix(A) \Rightarrow \pi \cdot x = x\big).$$

Definition 4.2 corresponds to [4, Definition 3.29], except that we write ρ there instead of Ω:

DEFINITION 4.2. Suppose X is a set with a permutation action over A, and $X \subseteq \mathsf{X}$ and $A \subseteq \mathsf{A}$. Then define $\Omega A.X$ by

$$\Omega A.X = \{\pi{\cdot}x \mid x \in X,\ \pi \in \mathit{fix}(A)\}.$$

LEMMA 4.3. *Suppose* X *is a set with a permutation action over* A*, and* $x{\in}\mathsf{X}$*,* $X{\subseteq}\mathsf{X}$ *and* $A{\subseteq}\mathsf{A}$*. Then:*

1. *If* $A\$X$ *then* $x \in X$ *if and only if* $\Omega A.\{x\} \subseteq X$*.*
2. $\Omega A.X$ *is the least* $X \subseteq \Omega A.X \subseteq \mathsf{X}$ *such that* $A \$ \Omega A.X$*.*

4.2 Uniformly $\omega{+}1$-partitioned sets

DEFINITION 4.4. Suppose Λ is an ordinal and $\#\Lambda \leq \beth(\omega{+}1)$.

1. Call a partition P of a set X **uniform** when $\#p = \#p'$ for every $p, p' \in P$ (so every class has equal cardinality).
2. Call a set A equipped with a uniform Λ-ordered partition $P = \{p_\lambda \mid 0 \leq \lambda < \Lambda\}$ a **uniformly Λ-partitioned** set.
3. Given a uniformly Λ-partitioned set A, for each $1 \leq \lambda \leq \Lambda$ define $\mathsf{A}_\lambda \subseteq \mathsf{A}$ by

$$\mathsf{A}_\lambda = \bigcup\{p_{\lambda'} \mid 0 \leq \lambda' < \lambda\}.$$

So $\{\mathsf{A}_\lambda \mid 1 \leq \lambda < \Lambda\}$ is an increasing sequence of subsets of A, all of which can be bijected with one another and with A. We will only care about the case $\Lambda = \omega{+}1$ henceforth.

DEFINITION 4.5. Suppose A and A' are uniformly $\omega{+}1$-partitioned sets. Then a **bijection** $\sigma : \mathsf{A} \cong \mathsf{A}'$ of A and A' as uniformly $\omega{+}1$-partitioned sets is a bijection on underlying sets that respects the partition in the sense that $\forall\lambda{\leq}\omega{+}1.(a \in \mathsf{A}_\lambda \Leftrightarrow \sigma(a) \in \mathsf{A}'_\lambda)$. This can be simplified to

$$\forall\lambda{<}\omega.(a \in \mathsf{A}_\lambda \Leftrightarrow \sigma(a) \in \mathsf{A}'_\lambda).$$

(Or $\forall\lambda, a.(a \in P_\lambda \Leftrightarrow a \in P'_\lambda)$, where P is the uniform partition of A and P' is that of A'.)

NOTATION 4.6. Suppose X is a set with a permutation action over a uniformly $\omega{+}1$-partitioned set A and $\lambda \leq \omega{+}1$. Define X_λ and $\mathsf{X}_\exists$ by

$$\mathsf{X}_\lambda = \{x \in \mathsf{X} \mid \mathsf{A}_\lambda \$ x\} \quad \text{and} \quad \mathsf{X}_\exists = \textstyle\bigcup_{\lambda<\omega} \mathsf{X}_\lambda.$$

Note that $\mathsf{X}_{\omega+1} = \mathsf{X}$, and that $\mathsf{X}_\omega \neq \mathsf{X}_\exists$. A special case of Notation 4.6 is of particular interest:

REMARK 4.7. Suppose A is a uniformly $\omega{+}1$-partitioned set. Then $\mathit{Perm}(\mathsf{A})$ naturally inherits a **functional** permutation action $(\pi{\cdot}\pi')(a) = \pi(\pi'(\pi^{-1}(a)))$. Then we can check that

$$\forall\lambda{<}\omega.\mathit{Perm}(\mathsf{A})_\lambda = \mathit{moves}(\mathsf{A}_\lambda) \quad \text{and} \quad \mathit{Perm}(\mathsf{A})_\exists = \textstyle\bigcup_{\lambda<\omega} \mathit{moves}(\mathsf{A}_\lambda).$$

4.3 The category of large uniformly $\omega{+}1$-partitioned sets

NOTATION 4.8. Suppose A and A' are sets and $\sigma \in \mathsf{A} \cong \mathsf{A}'$ is a bijection and $\pi \in \mathit{Perm}(\mathsf{A})$ is a permutation of A. Then define $\sigma(\pi)$ in the natural way such that

$$\sigma(\pi) : \sigma(a) \longmapsto \sigma(\pi(a)).$$

DEFINITION 4.9. Write PART for the category defined by:

- Objects $S \in \mathsf{PART}$ are tuples $(\mathsf{A}, \mathsf{X}, \cdot)$ where
  - $\mathsf{A} = \mathit{atm}(S)$ is a large uniformly $\omega{+}1$-partitioned set, which we call the **atoms** of S.
  - $\mathsf{X} = \mathit{elt}(S)$ is a large set, which we call the **elements** of S.

- $\cdot$ is a permutation action of A on X and X has small partition support over A (Definition 3.9(2)).

- Arrows $\sigma : S \to T \in \mathsf{PART}$ are pairs of maps

$$\begin{aligned} atm(\sigma) &\in atm(S) \to atm(T) \quad \text{and} \\ elt(\sigma) &\in elt(S)_{\exists} \to elt(T)_{\exists} \quad \text{such that:} \end{aligned}$$

1. $atm(\sigma)$ is a bijection of A and A' as uniformly ω+1-partitioned sets (Definition 4.5).
2. $elt(\sigma)$ is **bounded equivariant**, meaning that for every $\pi \in Perm(atm(S))_{\exists}$ (Remark 4.7) and $x \in elt(S)_{\exists}$ we have

$$elt(\sigma)(\pi{\cdot}x) = atm(\sigma)(\pi){\cdot}elt(\sigma)(x), \quad \text{or just} \quad \sigma(\pi{\cdot}x) = \sigma(\pi){\cdot}\sigma(x)$$

   if we abuse notation and write σ for both $atm(\sigma)$ and $elt(\sigma)$; $atm(\sigma)(\pi)$ is from Notation 4.8 and maps $atm(\sigma)(a)$ to $atm(\sigma)(\pi(a))$ for every $a \in \mathsf{A}$.

REMARK 4.10. A reasonable generalisation of Definition 4.9 would be to allow $atm(\sigma)$ to be an injection, but not necessarily a bijection. This generality is not needed for the ConNF proof.

On the other hand, we *will* find it useful to allow the generality of permitting S and S' to have different sets of atoms, even if they can be bijected.

LEMMA 4.11. *Suppose* $\sigma : S \to T \in \mathsf{PART}$. *Then* $elt(\sigma) \in elt(S)_{\exists} \to elt(T)_{\exists}$ *is* **uniformly partitioned**, *meaning that for every* $\lambda < \omega$ *we have*

$$x \in elt(S)_\lambda \quad \textit{implies} \quad elt(\sigma)(x) \in elt(T)_\lambda$$

and thus for every $\lambda < \omega$ *the map* $elt(\sigma)$ *restricts to a map in* $elt(S)_\lambda \to elt(T)_\lambda$.

Proof. By calculations using equivariance of σ (condition 2 in Definition 4.9), since if $\pi{\cdot}x = x$ then $elt(\sigma)(x) = elt(\sigma)(\pi{\cdot}x) = \pi{\cdot}elt(\sigma)(x)$. ◀

4.4 The power object functor

DEFINITION 4.12. Suppose X is a set with a permutation action over A. Define $pow_{sps}(\mathsf{X})$ by

$$pow_{sps}(\mathsf{X}) = \{X \subseteq \mathsf{X} \mid X \text{ has small partition support over } \mathsf{A}\}$$

where X has the *pointwise* permutation action (Definition 3.11) $\pi{\cdot}X = \{\pi{\cdot}x \mid x \in X\}$.

DEFINITION 4.13. Given $S \in \mathsf{PART}$ write $\mathsf{Po}(S) \in \mathsf{PART}$ for the **power object** defined by:
1. $atm(\mathsf{Po}(S)) = atm(S)$.
2. $elt(\mathsf{Po}(S)) = pow_{sps}(elt(S))$.

REMARK 4.14. We see from condition 1 of Definition 4.13 that S and $\mathsf{Po}(S)$ use the same set of atoms. Intuitively, the power object $\mathsf{Po}(\sigma)$ takes a small-partition-supported powerset of $elt(S)$.

DEFINITION 4.15. Suppose $S{=}(\mathsf{A}, \mathsf{X}, \cdot)$, $S'{=}(\mathsf{A}', \mathsf{X}', \cdot) \in \mathsf{PART}$ and $\sigma : S \to S' \in \mathsf{PART}$.

Then define $\mathsf{Po}(\sigma) : \mathsf{Po}(S) \to \mathsf{Po}(S')$ as follows:

- If $a \in \mathsf{A}$ then $\mathsf{Po}(\sigma)(a) = \sigma(a)$.
- If $\lambda < \omega$ and $X \in pow_{sps}(\mathsf{X})_\lambda$ then define (Ω is from Definition 4.2)

$$\mathsf{Po}(\sigma)(X) = \Omega A'_\lambda.\{\sigma(x) \mid x \in X_{\exists}\}.$$

REMARK 4.16. It is not obvious that $\mathsf{Po}(\sigma)(X)$ is well-defined or a functor. If $X \in pow_{sps}(\mathsf{X})_\lambda$ for $\lambda < \omega$ then $X \in pow_{sps}(\mathsf{X})_{\lambda+1}$; does λ affect the value computed by $\mathsf{Po}(\sigma)$? It does not; but this requires proof, which we consider next culminating with Proposition 4.20 and Corollary 4.21.

4.5 Three remarkable and useful facts: the squashing lemmas

It turns out that the notion of being uniformly ω+1-partitioned from Subsection 4.2 combines fruitfully with the notion of small partition support from Subsection 3.2, and we obtain three remarkable and useful facts which we will call collectively the *squashing lemmas*:

- Lemma 4.17 (and its corollary Corollary 4.18),
- Proposition 4.19, and
- Proposition 4.20.

LEMMA 4.17. *Suppose* $(\mathsf{A}, \mathsf{X}, \cdot) \in \mathsf{PART}$. *Then for every* $x \in \mathsf{X}$ *and* $\lambda < \omega$ *there exists a* **squashing permutation** $\pi \in Perm(\mathsf{A})$ *such that*

$$\pi \in \mathit{fix}(\mathsf{A}_\lambda) \quad \textit{and} \quad \mathsf{A}_{\lambda+1}\$\pi{\cdot}x.$$

Proof. By assumption there exists a small partition $E \in part_s(\mathsf{A})$ such that $E\$X$. We now unroll definitions and use facts of cardinalities; details in [4, Theorem 5.20] (the first squashing property). ◀

COROLLARY 4.18. *Suppose* $(\mathsf{A}, \mathsf{X}, \cdot) \in \mathsf{PART}$ *and* $\lambda < \omega$ *and* $X, X' \in pow_{sps}(\mathsf{X})_\lambda$. *Then if* $X \neq X'$ *then there exists* $x \in \mathsf{X}_{\lambda+1}$ *such that either* $x \in X \wedge x \notin X'$ *or* $x \notin X \wedge x \in X'$.

Proof. By sets extensionality if $X \neq X'$ then there exists $x \in \mathsf{X}$ such that $x \in X \wedge x \notin X'$ or $x \notin X \wedge x \in X'$. What we do not know is whether $\mathsf{A}_{\lambda+1}\$x$ is possible.

Assume $x \in X \wedge x \notin X'$; the other case is similar. By assumption there exists $E \in part_s(\mathsf{A})$ with $E\$x$. From Lemma 4.17 we obtain $\pi \in \mathit{fix}(\mathsf{A}_\lambda)$ such that $\mathsf{A}_{\lambda+1}\$\pi{\cdot}x$.

Now $\pi \in \mathit{fix}(\mathsf{A}_\lambda)$ and $\mathsf{A}_\lambda\$X, X'$ and it follows that $\pi{\cdot}X = X$ and $\pi{\cdot}X' = X'$, and so by the pointwise action that $\pi{\cdot}x \in X \wedge \pi{\cdot}x \notin X'$. So $\pi{\cdot}x$ is the element we need. ◀

PROPOSITION 4.19. *Suppose* $(\mathsf{A}, \mathsf{X}, \cdot) \in \mathsf{PART}$ *and* $\pi \in Perm(\mathsf{A})$ *and* $\lambda < \omega$. *Then there exists* $\pi' \in Perm(\mathsf{A})_{\lambda+1}$ *such that:*

1. $\mathit{fix}(\pi) \subseteq \mathit{fix}(\pi')$ *(so* $\pi(a) = a$ *implies* $\pi'(a) = a$*).*
2. *For every* $x \in \mathsf{X}_\lambda$, $\pi'{\cdot}x = x$ *if and only if* $\pi{\cdot}x = x$.

Proof. Consider an orbit O of π. If O is finite, write it as a finite tuple and using Lemma 4.17 squash its support into $\mathsf{A}_{\lambda+1}$. If O is infinite, it is a $\mathbb{Z}$-tuple of elements; choose $x \in O$ and using Lemma 4.17 squash the support of each $x' \in O$ into $\mathsf{A}_{\lambda+1+i}$ where i is the distance of x' from x in O (so if $x' = x$ then $i = 0$). We obtain a $\mathbb{Z}$-tuple supported by A_ω, and since $\mathsf{A} \setminus \mathsf{A}_\omega$ is large we can squash to obtain a tuple supported by $\mathsf{A}_{\lambda+1}$. See [4, Proposition 5.24] (the second squashing property). ◀

PROPOSITION 4.20. *Suppose* $S = (\mathsf{A}, \mathsf{X}, \cdot) \in \mathsf{PART}$ *and* $S' = (\mathsf{A}', \mathsf{X}', \cdot) \in \mathsf{PART}$ *and* $\sigma : S \to S' \in \mathsf{PART}$. *Consider* $\lambda < \omega$ *and suppose* $X \in pow_{sps}(\mathsf{X})_\lambda$. *Then for any* $\lambda \leq \mu < \omega$ *(Definition 4.15 is the case* $\mu = \lambda$*)*

$$\mathsf{Po}(\sigma)(X) = \Omega A'_\mu.\{\sigma(x) \mid x \in X_\exists\}.$$

As a corollary, Definition 4.15 is well-defined.

Proof. By assumption $atm(\sigma) \in \mathsf{A} \cong \mathsf{A}'$ so for brevity just assume $\mathsf{A} = \mathsf{A}'$. By construction $\mathsf{A}_\lambda \subseteq \mathsf{A}_\mu$ and it follows from Definition 4.2 that $\Omega A_\mu.\{\sigma(x) \mid x \in X_\exists\} \subseteq \Omega A_\lambda.\{\sigma(x) \mid x \in X_\exists\}$.

So it would suffice to prove that if $\pi \in \mathit{fix}(\mathsf{A}_\lambda)$ then π fixes $\Omega A_\mu.\{\sigma(x) \mid x \in X_\exists\}$. By Proposition 4.19 we may assume without loss of generality that $\pi \in \mathit{fix}(\mathsf{A}_\lambda)$ and $\pi \in Perm(\mathsf{A})_{\mu+1} \subseteq Perm(\mathsf{A})_\exists$. Note that it follows by facts about permutations that if $\pi \in Perm(\mathsf{A})_\exists$ then $\pi{\cdot}X_\exists = X_\exists$ (meaning that $x \in X_\exists \Leftrightarrow \pi{\cdot}x \in X_\exists$ for any $x \in \mathsf{X}$).

Furthermore $\mathsf{A}_\mu \subseteq \mathsf{A}_\lambda$ so it would suffice that π fixes $\{\sigma(x) \mid x \in X_\exists\}$. But this is not hard:

$$\begin{aligned}
\pi{\cdot}\{\sigma(x) \mid x \in X_\exists\} &= \{\pi{\cdot}\sigma(x) \mid x \in X_\exists\} && \text{Pointwise action}\\
&= \{\sigma(\pi{\cdot}x) \mid x \in X_\exists\} && \sigma \text{ bounded equivariant}\\
&= \{\sigma(x) \mid x \in \pi{\cdot}(X_\exists)\} && \text{Fact}\\
&= \{\sigma(x) \mid x \in X_\exists\} && \pi \in \mathsf{Perm}(\mathsf{A})_\exists.
\end{aligned}$$

So we are done. For more details see [4, Lemma 6.8(2) & 6.16(2)]. ◀

COROLLARY 4.21. Po *from Definition 4.15 is a functor.*

Proof. Po clearly maps the identity to itself. Now suppose $\sigma : S \to S'$ and $\sigma' : S' \to S''$ and write $\mathsf{X} = elt(S)$ and consider some $X \in pow_{sps}(\mathsf{X})_\exists$. So $X \in pow_{sps}(\mathsf{X})_\lambda$ for some $\lambda < \omega$ and using Proposition 4.20 it can be checked that it suffices to verify that

$$\{\sigma'(x') \mid x' \in \{\sigma(x) \mid x \in \mathsf{X}_\exists\}\} = \{\sigma'(\sigma(x)) \mid x \in \mathsf{X}_\exists\}.$$

This is a fact of function composition, so we are done. ◀

4.6 More on uniformly partitioned sets

We further apply the squashing lemmas to prove the nontrivial fact that Po preserves $=$, $\in$, and $\rightarrowtail$:

NOTATION 4.22. Write $\sigma : S \rightarrowtail S' \in \mathsf{PART}$ when:

- $\sigma : S \to S' \in \mathsf{PART}$ and
- $elt(\sigma)$ is an injective map from $elt(S)_\exists$ to $elt(S')_\exists$ (Notation 4.6).

We call $\sigma : S \rightarrowtail S'$ a (uniformly partitioned) **injection**.

If $elt(\sigma)$ is a bijection and not just an injection, then we call σ a **bijection**. Note that a 'bijective' σ is partial in the sense that it bijects $elt(S)_\exists$ with $elt(S')_\exists$, not $elt(S)$ with $elt(S')$. This is why it is not obvious that injectivity is preserved by Po.

THEOREM 4.23. *Suppose* $S = (\mathsf{A}, \mathsf{X}, \cdot)$ *and* $S' = (\mathsf{A}', \mathsf{X}', \cdot')$ *and* $\sigma : S \rightarrowtail S' \in \mathsf{PART}$. *Then:*

1. *If* $\lambda < \omega$ *and* $X \in pow_{sps}(\mathsf{X})_\lambda$ *(Notation 4.6(1)) and* $x \in \mathsf{X}_{\lambda+1}$ *then*
$$x \in X \quad \textit{if and only if} \quad \sigma(x) \in \mathsf{Po}(\sigma)(X).$$
2. *If* $X, Y \in pow_{sps}(\mathsf{X})_\lambda$ *then*
$$\mathsf{Po}(\sigma)(X) = \mathsf{Po}(\sigma)(Y) \quad \textit{implies} \quad X = Y.$$
3. $\mathsf{Po}(\sigma)$ *is bounded equivariant, meaning that for every* $\pi \in Perm(\mathsf{A})_\exists$ *(Remark 4.7) and every* $X \in pow_{sps}(\mathsf{X})_\exists$ *we have*
$$\mathsf{Po}(\sigma)(\pi{\cdot}x) = \pi{\cdot}\mathsf{Po}(\sigma)(x).$$

As a corollary, $\sigma : S \rightarrowtail S' \in \mathsf{PART}$ *implies* $\mathsf{Po}(\sigma) : \mathsf{Po}(S) \rightarrowtail \mathsf{Po}(S') \in \mathsf{PART}$.

Proof. **1.** If $\sigma(x){\in}\sigma(X)$ then $\sigma(x) = \pi'{\cdot}\sigma(y)$ for some $y{\in}\mathsf{X}$ such that $\mathsf{A}_{\lambda+1}\$y$ and $\pi'{\in}fix(\mathsf{A}'_\lambda)$. Using Proposition 4.19, assume without loss of generality that π' permutes only elements in $\mathsf{A}'_{\lambda+2} \setminus \mathsf{A}'_\lambda$. Since σ is bijective on atoms, we may write $\pi' = \sigma(\pi)$ (Notation 4.8). Using Proposition 4.20, $\pi'{\cdot}\sigma(y) = \sigma(\pi{\cdot}y)$.
By equivariance $\sigma(x) = \sigma(\pi{\cdot}y)$, and σ is injective on underlying sets so that $x = \pi{\cdot}y$.
Now recall that $y \in X$, and $\mathsf{A}_\lambda\$X$ and $\pi \in fix(\mathsf{A}_\lambda)$ so that $X = \pi{\cdot}X$. It follows from the pointwise action that $x = \pi{\cdot}y \in X$.

2. If $X, Y \in pow_{sps}(\mathsf{X})_\lambda$ and $X \neq Y$ then by Corollary 4.18 there exists $x \in pow_{sps}(\mathsf{X})_{\lambda+1}$ such that $x \in X \wedge x \notin Y$ or $x \notin X \wedge x \in Y$. By part 1 of this result $\sigma(X) \neq \sigma(Y)$ as required.

3. Bounded equivariance follows by routine calculations using Proposition 4.20 and choosing μ' such that $\pi \in moves(\mathsf{A}_{\mu'})$ (see also [4, Lemma 6.29]). ◀

$$\vDash \bot \Leftrightarrow \bot \qquad \vDash \phi \mathsf{\wedge} \phi' \Leftrightarrow \vDash \phi \wedge \vDash \phi' \qquad \vDash \neg\phi \Leftrightarrow \nvDash \phi$$
$$\vDash y \mathsf{\in} x \Leftrightarrow y \in x \qquad \vDash \exists a.\phi \Leftrightarrow \exists x \mathsf{\in} \mathsf{V}_{level(a)}.(\vDash \phi[a{:=}x])$$

Figure 1 The denotation

5 TST+ is consistent

DEFINITION 5.1. For convenience extend term syntax with $\bigcup_{i\geq 1} \mathsf{V}_i$ (Definition 3.17) as constants, where if $x \in \mathsf{V}_i$ then $level(x) = i$. We give the TST+ syntax from Definition 2.1 a denotation in the powersets hierarchy from Definition 3.17 as per Figure 1. In that figure, we assume that ϕ is closed (no free variables, but possibly some constants).

DEFINITION 5.2. **1.** Suppose $i \geq 1$ and $level(a) = i$. If ϕ has (at most) one free variable a and is in quantifier normal form[4] then call ϕ **a-comprehensive**.

2. Define a mapping $\Gamma a.\phi$ from a-comprehensive ϕ to $\Gamma a.\phi \in \bigcup_{n\geq 1} pow^n(\mathsf{V}_i)$ by:

$$\begin{array}{rll} \Gamma a.\phi = \{x \in \mathsf{V}_i \mid \vDash \phi[a{:=}x]\} & & \phi \text{ is quantifier-free} \\ \Gamma a.(\forall b.\phi) = \{\Gamma a.(\phi[b{:=}y]) \mid y \in \mathsf{V}_j\} & & level(b) = j \\ \Gamma a.(\exists b.\phi) = \{\Gamma a.(\phi[b{:=}y]) \mid y \in \mathsf{V}_j\} & & level(b) = j \end{array}$$

(So $\forall b.\phi$ and $\exists b.\phi$ map to the same set.)

LEMMA 5.3. *Suppose $1 \leq j < i$ and $a \in \mathsf{Var}_i$ and ϕ is a-comprehensive with a quantifier prefix of length $n \geq 0$. Suppose $P \in \mathsf{EQ}_j$ and suppose $P\$x$ for every constant x mentioned in ϕ.*[5] *Then*

$$P \,\$\, \Gamma a.\phi.$$

Proof. By calculations checking $\Gamma a.\phi$ inherits the support of the constants in ϕ [4, Lemma 4.18]. ◀

LEMMA 5.4. *Suppose $1 \leq j \leq i$ and $n \geq 1$ and $X \in pow^n(\mathsf{V}_i)$ and $P \in \mathsf{EQ}_j$ and $P\$X$. Then $P\$\bigcap X$ and $P\$\bigcup X$.*

Proof. It is easy to check that if $\pi \cdot X = X$ then also $\pi \cdot \bigcap X = \bigcap X$ and $\pi \cdot \bigcup X = \bigcup X$. ◀

PROPOSITION 5.5. *Suppose ϕ has constants $\{y_1, \ldots, y_n\}$ and just one free variable a where $level(a) = i \geq 1$. Then $\{x \in \mathsf{V}_i \mid \vDash \phi[a{:=}x]\} \in \mathsf{V}_{i+1}$, and so the denotation in Figure 1 satisfies comprehension.*

Proof. Assume without loss of generality that ϕ is in quantifier normal form. By Lemma 5.3 $\Gamma a.\phi$ has small partition support at every $0 \leq j \leq i$. We recover $\{x \in \mathsf{V}_i \mid \vDash \phi[a{:=}x]\}$ with intersections $\bigcap$ and unions $\bigcup$ following the quantifier prefix of ϕ. By Lemma 5.4, $\{x \in \mathsf{V}_i \mid \vDash \phi[a{:=}x]\} \in \mathsf{V}_{i+1}$ ◀

REMARK 5.6. Proposition 5.5 uses the 'at every lower level' part of the support conditions in Definition 3.17. If we did not insist on small partition support at *every* lower level, then a comprehension using constants from different levels might lose its overall symmetry.

THEOREM 5.7. *TST+ is consistent.*

Proof. The hierarchy $(\mathsf{V}_i \mid i \geq 1)$ from Definition 3.17 is a model: Extensionality is a fact of sets. Comprehension is Proposition 5.5. For typical ambiguity, we reason as follows:

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[4] $\phi = \mathsf{Q}.\phi'$ where ϕ' is quantifier-free and Q represents a $\forall\exists$-quantifier prefix. ϕ may mention constants.

[5] It is a fact that if $P_1, P_2 \in \mathsf{EQ}_j$ then there is a partition $P_1 \wedge P_2 \in \mathsf{EQ}_j$ that is their greatest lower bound. So we can always find a suitable P.

1. By Theorem 3.18 we have a bijection $\sigma : \mathsf{V}_2 \cong \mathsf{V}_1$.
2. We choose some uniform ω+1-partition on V_1, and using it we present V_1 as an object in PART.
3. We transfer this partition along σ to V_2, and present V_2 as an object in PART.
4. We obtain a bijective arrow $\sigma : \mathsf{V}_2 \rightarrowtail \mathsf{V}_1 \in \mathsf{PART}$.
   'σ bijective' here means something specific, and σ is partial on underlying sets in a very specific way (Notation 4.22); this partiality is what helps sidestep diagonalisation counterarguments.
5. Using Theorem 4.23 we repeat and extend this to a *sequence* of bijective arrows in PART between V_i and $\mathsf{V}_{i\text{-}1}$ for every $i \geq 2$.
6. By the first squashing lemma Lemma 4.17, if a witness $x \in \mathsf{V}_j$ exists to some $\exists a.\phi$ in the language of stratified sets — where $\exists a.\phi$ mentions only variables no lower than level 2; along with finitely many constants from $\bigcup_{i\geq 2} \mathsf{V}_i$ (so also no lower than level 2) and supported by $(\mathsf{V}_2)_\lambda$ for some $\lambda < \omega$ — then is also a witness $x \in (\mathsf{V}_j)_{\lambda+1}$, so that x it is in the domain of our bijections.
7. We are now essentially done, because we know that if an existential witness exists at a 'high' level, then one exists at a 'lower' level. Typical ambiguity follows by induction on predicates. ◀

REMARK 5.8. We have elided a technical detail: the hierarchy in Definition 3.17 imposes small partition support conditions on $X \in \mathsf{V}_i$ for *every* smaller i. We should check that σ preserves these symmetries. It does. The proof is by diagram-chasing computations, checking that permutations commute as they should. The relevant calculations are in Subsection 6.2.3 and Subsection 6.3 of [4].

6 Conclusions

ZF, HOL[6], and NF/TST+ are cousins: they arose contemporaneously in response to Russell's paradox.

If NF/TST+ has been less studied than ZF and HOL[7] this may partly be because ZF and HOL have cumulative hierarchy models (as featured in many an undergraduate course on mathematical foundations). NF and TST+ had no such model, and indeed no models were known at all, and indeed NF and TST+ were not known consistent. It makes sense why powerful computer science tools have so far been based on derivatives of ZF and HOL, but not NF.

We have studied NF using ideas from computer science in general and from nominal techniques in particular (as presented in e.g. [6, 10, 3]); and bringing to bear this combination of ideas along with new ones, we have proposed a model for TST+ (Definition 3.17). Furthermore, this is not just *any* model: it is a *cumulative hierarchy* model — like the ZF cumulative sets hierarchy model — and thus is a natural model for an extensional theory.

If the TST+ cumulative hierarchy from Definition 3.17 is more complex than the ZF sets hierarchy, with its small partition support conditions, well: this buys new bijections, limits cardinalities, and gives us typical ambiguity and a universal set,[8] and lets us model a stronger theory.

The tools described in this paper are hard, but in interesting ways: small partition support (Definition 3.9), the power object functor (Definition 4.15), and uniform partitions (Subsection 4.2) are new ideas, with good behaviour (like the squashing lemmas in Subsection 4.5, and Theorem 4.23) and they serve a nontrivial application (build a model of TST+ and so prove ConNF). Future work includes applying these ideas to devise more structured type theories with universal sets, including constructive ones; TST+ is the case of a classical theory in which types are just $\mathbb{N}_{\geq 1}$.

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[6] Simply-typed λ-calculus with booleans and natural numbers

[7] Example: the paper on confluence and strong normalisation of the language of stratified sets [5] answers basic rewriting-theory-flavoured questions about the syntax of NF and TST+, which arguably anyone with my background would ask.

[8] To be precise: TST+ has a universal set at each level; NF has a universal set.

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